

CHEMISTRY

Q.1

For the manufacture of ammonia by Haber's process,
i.e. $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g}); \Delta H = -22 \text{ kcal}$
the favourable conditions are

- (a) High temperature, high pressure
- (b) Low temperature, low pressure
- (c) High temperature, low pressure
- (d) Low temperature, high pressure

And: (d)

Q.2

Match the column I with column II and mark the appropriate choice.

Column I		Column II	
(A)	CH_3COONa	(i)	Almost neutral, $\text{pH} > 7$ or < 7
(B)	NH_4Cl	(ii)	Acidic, $\text{pH} < 7$
(C)	NaNO_3	(iii)	Alkaline, $\text{pH} > 7$
(D)	$\text{CH}_3\text{COONH}_4$	(iv)	Neutral, $\text{pH} = 7$

- (a) (A) $\rightarrow$ (i), (B) $\rightarrow$ (ii), (C) $\rightarrow$ (iii), (D) $\rightarrow$ (iv)
- (b) (A) $\rightarrow$ (ii), (B) $\rightarrow$ (iii), (C) $\rightarrow$ (iv), (D) $\rightarrow$ (i)
- (c) (A) $\rightarrow$ (iii), (B) $\rightarrow$ (ii), (C) $\rightarrow$ (iv), (D) $\rightarrow$ (i)
- (d) (A) $\rightarrow$ (iv), (B) $\rightarrow$ (i), (C) $\rightarrow$ (iii), (D) $\rightarrow$ (ii)

And: (c)

Q.3

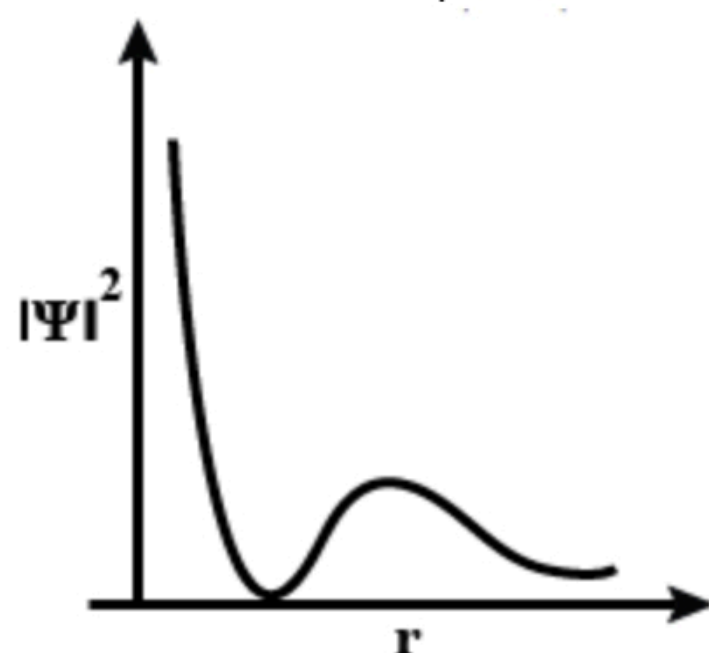
Amongst $\text{Ni}(\text{CO})_4$, $[\text{Ni}(\text{CN})_4]^{2-}$ and $[\text{NiCl}_4]^{2-}$

- (a) $\text{Ni}(\text{CO})_4$ and $[\text{Ni}(\text{CN})_4]^{2-}$ are diamagnetic and $\text{Ni}(\text{CO})_4$ is paramagnetic
- (b) $[\text{NiCl}_4]^{2-}$ and $[\text{Ni}(\text{CN})_4]^{2-}$ are diamagnetic and $\text{Ni}(\text{CO})_4$ is paramagnetic
- (c) $\text{Ni}(\text{CO})_4$ and $[\text{Ni}(\text{CN})_4]^{2-}$ are diamagnetic and $[\text{NiCl}_4]^{2-}$ is paramagnetic
- (d) $\text{Ni}(\text{CO})_4$ is diamagnetic and $[\text{NiCl}_4]^{2-}$ and $[\text{Ni}(\text{CN})_4]^{2-}$ are paramagnetic

Ans: (c)

Q.4

The graph between $|\Psi|^2$ and r (radial distance) is shown below. This represents:



- | | |
|----------------|----------------|
| (1) 3s orbital | (2) 1s orbital |
| (3) 2p orbital | (4) 2s orbital |

Ans:(4)

Q.5

Given below are two statements :

Statement-I : α and β forms of sulphur can change reversibly between themselves with slow heating or slow cooling.

Statement-II : At room temperature the stable crystalline form of sulphur is monoclinic sulphur.

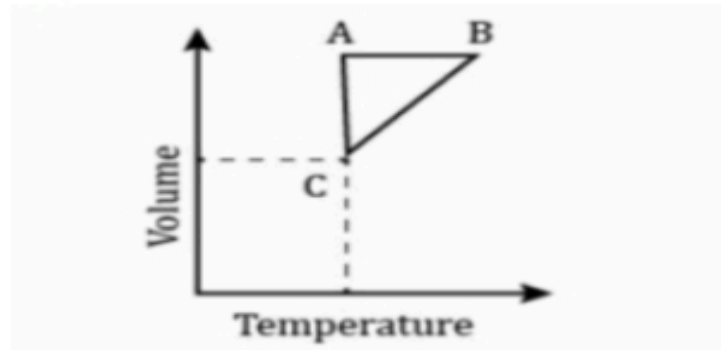
In the light of the above statements, choose the correct answer from the options given below:

- (1) Statement I is false but Statement II is true.
- (2) Both Statement I and Statement II are true.
- (3) ~~Both~~ Statement I is true but Statement II is false.
- (4) Both Statement I and Statement II are false.

Ans:(3)

Q.6

Five moles of a gas is put through a series of changes as shown graphically in a cyclic process. The $A \rightarrow B$, $B \rightarrow C$, and $C \rightarrow A$ respectively are:



- (a) Isochoric, Isobaric, Isothermal
- (b) Isobaric, Isochoric, Isothermal
- (c) Isothermal, Isobaric, Isochoric
- (d) Isochoric, Isothermal, Isobaric

Ans: (1)

Q.7

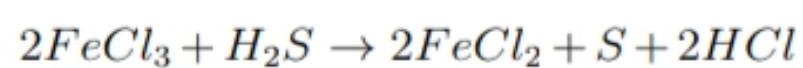
If a reaction follows the Arrhenius equation, the plot in k vs $\frac{1}{RT}$ gives a straight line with a gradient $-\gamma$ unit. The energy required to activate the reactant is:

- (a) $\frac{\gamma}{R}$ unit
- (b) γR unit
- (c) γ unit
- (d) $-\gamma$ unit

Ans: (c)

Q.8

In the reaction:



- (a) $FeCl_3$ acts as oxidising agent
- (b) Both H_2S and $FeCl_2$ are oxidised
- (c) $FeCl_2$ is oxidised while H_2S is reduced
- (d) $FeCl_2$ is oxidised while H_2S is reduced
- (e) H_2S acts as oxidising agent

Ans: (1)

Q.9

Pure N_2 can be obtained by heating _____.

- A. NH_3 with CuO
- B. NH_4NO_3
- C. $(NH_4)_2Cr_2O_7$
- D. $Ba(N_3)_2$

Ans: (d)

Q.10

Given below are two statements:

Statement I: Group 13 trivalent halides get easily hydrolyzed by water due to their covalent nature.

Statement II: AlCl_3 upon hydrolysis in acidified aqueous solution forms octahedral $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$ ion.

In the light of the above statements, choose the correct answer from the options given below:

- 1) Statement I is false but Statement II is true
- 2) Both Statement I and Statement II are true
- 3) Both Statement I and Statement II are false
- 4) Statement I is true but Statement II is false

Ans: (2)

Q.11

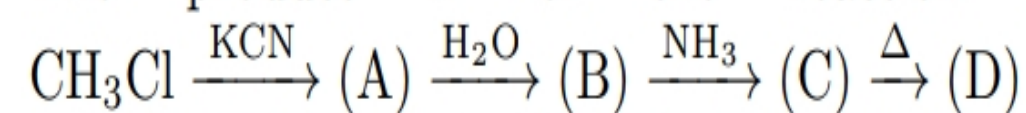
The brown ring test for NO_2^- and NO_3^- is due to the formation of complex ion with a formula:

- 1) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
- 2) $[\text{Fe}(\text{NO})(\text{CN})_5]^{3-}$
- 3) $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]^{2+}$
- 4) $[\text{Fe}(\text{H}_2\text{O})(\text{NO})]^{3+}$

Ans: (3)

Q.12

The product D of the reaction is?

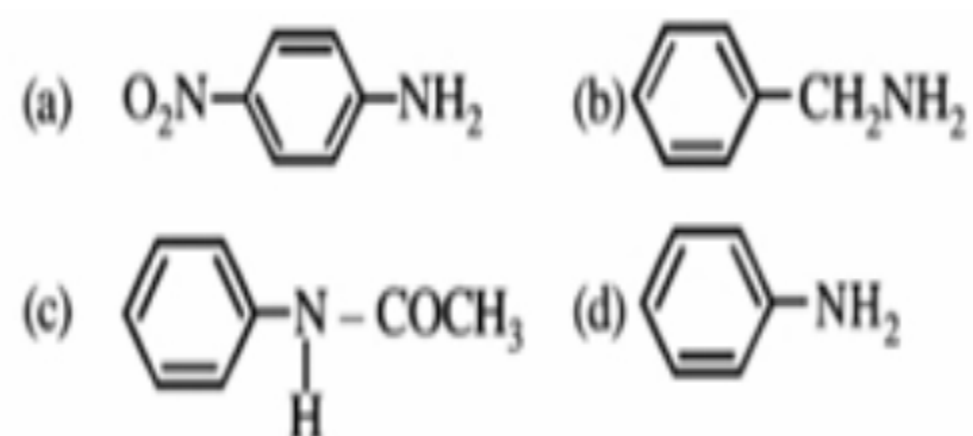


1. $\text{CH}_3\text{CH}_2\text{NH}_2$
2. CH_3CN
3. HCONH_2
4. CH_3CONH_2

Ans: (4)

Q.13

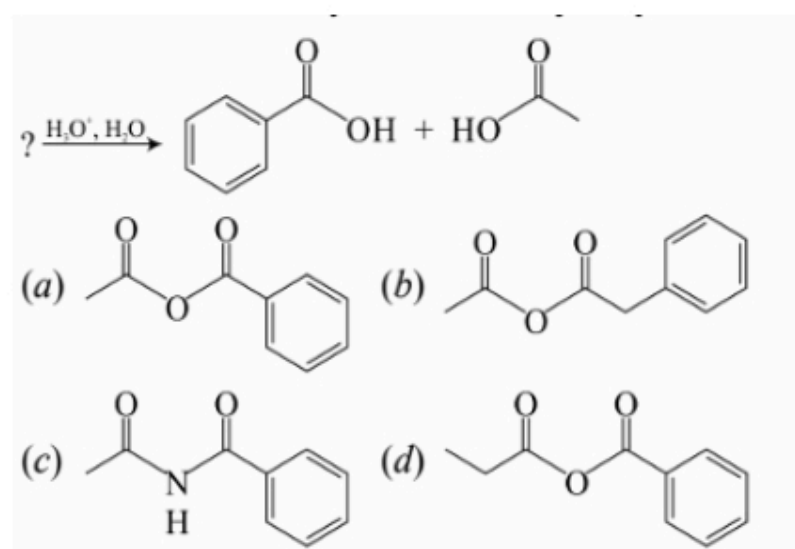
Which of the following compounds is most basic?



Ans: (b)

Q.14

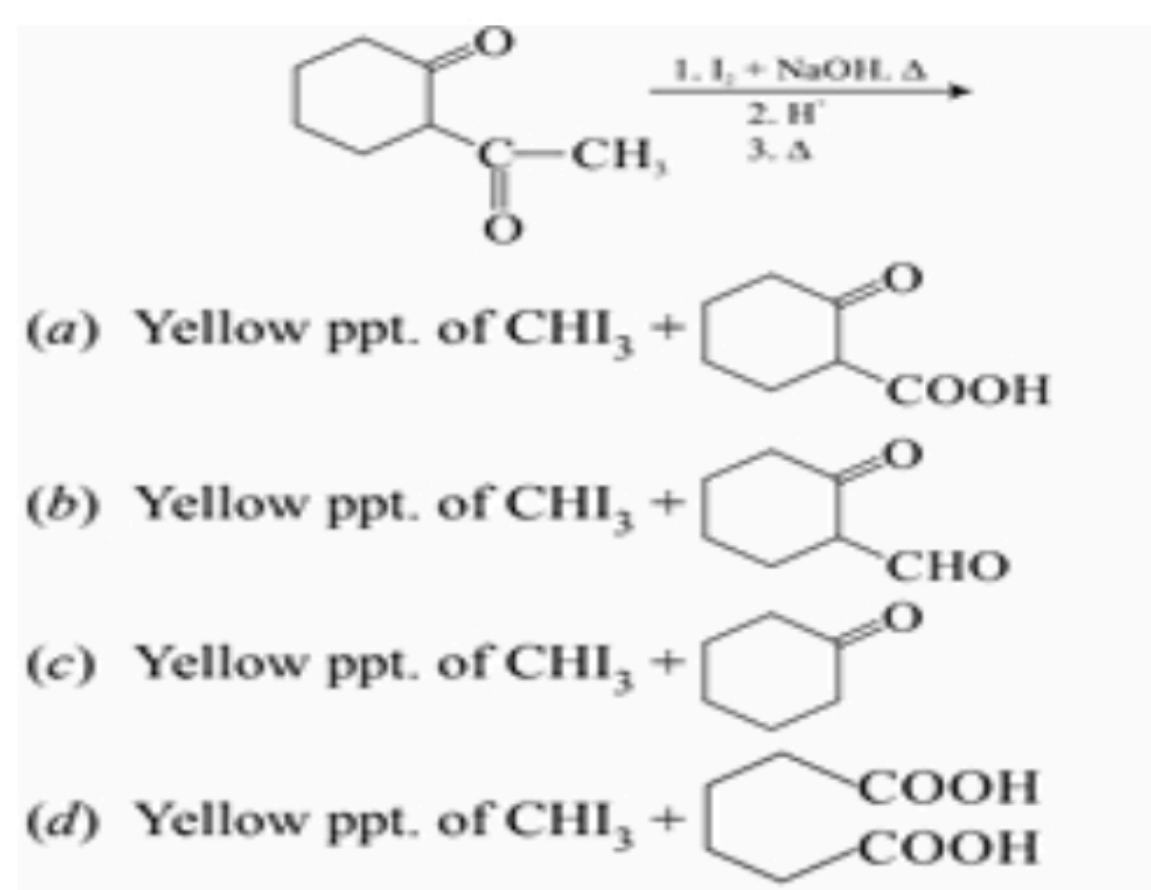
What is the acid anhydride that is hydrolyzed?



Ans: (1)

Q.15

End products of the following sequence of reactions are



Ans: (c)

Q.16

The ether that undergoes electrophilic substitution reactions is

1. $\text{CH}_3\text{OC}_2\text{H}_5$
2. $\text{C}_6\text{H}_5\text{OCH}_3$
3. CH_3OCH_3
4. $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$

Ans: (2)

Q.17

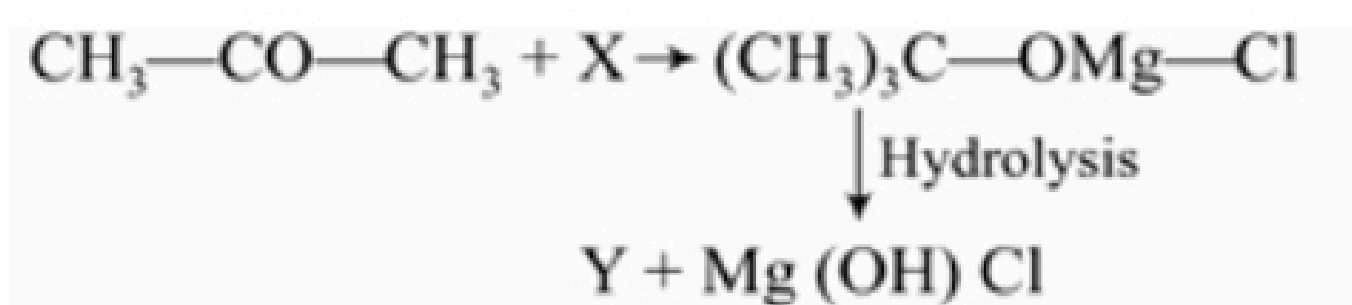
Compound 'A' reacts with PCl_5 to give 'B' which on treatment with KCN followed by hydrolysis gives propanoic acid as the product. What is 'A'?

1. Ethane
2. Propane
3. Ethyl chloride
4. Ethyl alcohol

Ans: (4)

Q.18

Identify the reactant X and the product Y

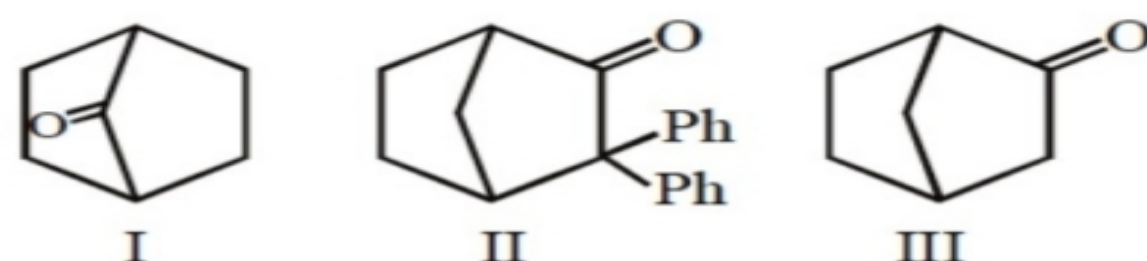


- (a) $\text{X} = \text{MgCl}_2$; $\text{Y} = \text{CH}_3\text{CH} = \text{CH}_2$
- (b) $\text{X} = \text{CH}_3\text{MgCl}$; $\text{Y} = \text{C}_2\text{H}_5\text{COCH}_3$
- (c) $\text{X} = \text{CH}_3\text{MgCl}$; $\text{Y} = (\text{CH}_3)_3\text{COH}$
- (d) $\text{X} = \text{C}_2\text{H}_5\text{MgCl}$; $\text{Y} = (\text{CH}_3)_3\text{COH}$

Ans: (c)

Q.19

Which among the given molecules can exhibit tautomerism?



- A) III Only
- B) Both I and III
- C) Both I and II
- D) Both II and III

Ans: (a)

Q.20

The hyper conjugative stabilities of tert-butyl cation and 2-butene, respectively, are due to:

- A. $\sigma \rightarrow p$ (empty) and $\sigma \rightarrow \pi^*$ electron delocalisations.
- B. $\sigma \rightarrow \sigma^*$ and $\sigma \rightarrow \pi$ electron delocalisations.
- C. $\sigma \rightarrow p$ (filled) and $\sigma \rightarrow \pi$ electron delocalisations.
- D. $p(\text{filled}) \rightarrow \sigma^*$ and $\sigma \rightarrow \pi^*$ electron delocalisations.

Ans: (a)